ECE 220 Computer Systems & Programming

Lecture 16 — Dynamic Memory Allocation

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One Function, Any Roster

One function reads a roster file — 5 students or 400 — and hands the records back to whoever called it.

? `read_roster(`*filename*`)`

it has to hand back **the records** and **how many** there are

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Where do those records live once `read_roster` returns?

Two Rosters, One Function

```
student *read_roster(char *filename, int *n) {
    student roster[200];   /* 200: fixed when compiled */
    FILE *fp = fopen(filename, "r");
    int i;
    if (fp == NULL)
        return NULL;
    fscanf(fp, "%d", n);
    for (i = 0; i < *n; i++)
        fscanf(fp, "%s %d %f", roster[i].Name,
               &roster[i].UIN, &roster[i].GPA);
    fclose(fp);
    return roster;          /* address of a local */
}
    sec_a = read_roster("roster.txt", &n_a);
    sec_b = read_roster("roster_b.txt", &n_b);
```

gcc: warning: address of stack memory associated with local variable 'roster' returned

Two Rosters, One Function

```
student *read_roster(char *filename, int *n) {
    student roster[200];   /* 200: fixed when compiled */
    FILE *fp = fopen(filename, "r");
    int i;
    if (fp == NULL)
        return NULL;
    fscanf(fp, "%d", n);
    for (i = 0; i < *n; i++)
        fscanf(fp, "%s %d %f", roster[i].Name,
               &roster[i].UIN, &roster[i].GPA);
    fclose(fp);
    return roster;          /* address of a local */
}
    sec_a = read_roster("roster.txt", &n_a);
    sec_b = read_roster("roster_b.txt", &n_b);
```

`roster.txt` holds 5 students, `roster_b.txt` holds 4. Section A prints

Draw the run-time stack (RTS) after the second call returns:



gcc: warning: address of stack memory associated with local variable 'roster' returned

`roster` lived in `read_roster`'s activation record (AR). The AR was popped, and the second call built its own in the same slots.

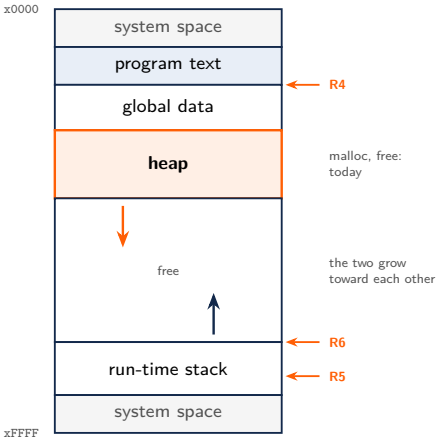
Today's Three Questions

The run-time stack lends a function memory and takes it back when the function returns.

1. Where does memory come from when its size is known only at run time, and the block must outlive the function that asked for it?
2. How does a program ask for a block, and how does it give it back?
3. What goes wrong when it forgets to give it back, gives it back twice, or keeps using it afterwards?

Lecture 5's memory map had a box marked "heap (Lecture 16)". Today fills it in.

Review: Where Things Live



Lower addresses at the top, as in Lecture 8.

The `student` struct: `Name[100]`, `UIN`, `GPA`;
`sizeof(student)` is 108 bytes.

- `student ece220[200];` outside `main`: 21600 bytes in _____, alive _____
- `student s1;` in `main`: 108 bytes in _____, popped when _____
- `student *s_ptr = ece220;` in `main`: the pointer is in _____; the address it holds is in _____
- `student roster[200];` in `read_roster`: 21600 bytes in _____, popped at return
- Today: a block from `malloc` is in the _____ until the program frees it

The heap and the stack grow toward each other through the free region between them.

"Static" vs Dynamic Allocation

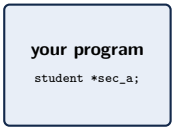
"Static" here means fixed by a declaration. It covers both storage classes: static (globals) and automatic (locals).

	"Static"	Dynamic
Mechanism		
Lifetime		
Location		
Size		

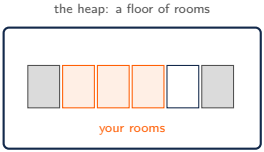
A dynamic block has no name. The only way back to it is the address `malloc` returned.

The Heap as a Hotel

Three parties: your program, the allocator at the front desk, and the rooms themselves.

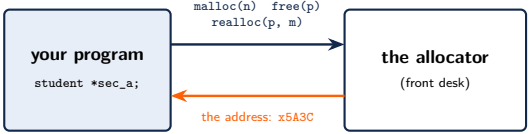


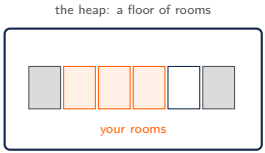




The Heap as a Hotel

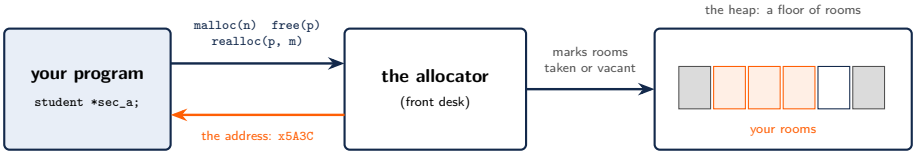
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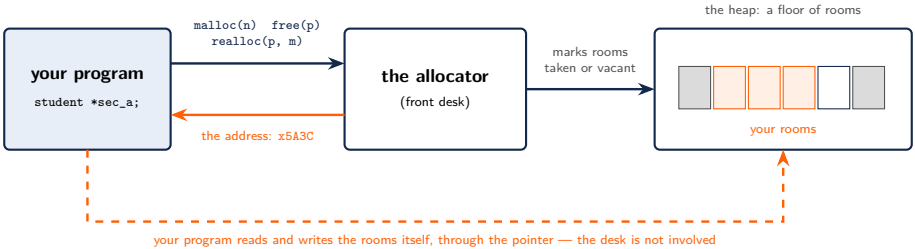
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The Heap as a Hotel

Three parties: your program, the allocator at the front desk, and the rooms themselves.



The desk hears from you only at `malloc`, `free` and `realloc`. In between, that one address is your only way back into the rooms.

malloc and free

```
ptr = (int *)malloc(sizeof(int));
if (ptr == NULL) {
    printf("ERROR - malloc failure!\n");
    return 1;
}
*ptr = 10;
printf("*ptr = %d\n", *ptr);
free(ptr);
```

Output: `*ptr = 10`

An array of 10 ints, in one block:

`arr =` _____

Then `arr[i]` as for any array, and one `free(arr)` returns all _____ bytes.

`void free(void *ptr)`

- gives the whole block back; `ptr` must be an address that `malloc`, `calloc` or `realloc` returned

`void *malloc(size_t size)`

- asks the allocator for `size` contiguous bytes on the heap; `size_t` is an unsigned integer type
- returns the block's first address as a `void *`, an address with no type; the cast `(int *)` gives it one
- returns `NULL` when no block that large is free. Writing through `NULL` is a _____
- the bytes are _____

In-Class Problem: read_roster on the Heap

```
student *read_roster(char *filename, int *n) {
    student *roster;
    fscanf(fp, "%d", n);
    /* a block of *n students; NULL if none */




    for (i = 0; i < *n; i++)
        fscanf(fp, "%s %d %f", roster[i].Name,
               &roster[i].UIN, &roster[i].GPA);
    fclose(fp);
    return roster;   /* heap address: valid */
}
```

and in `main`:

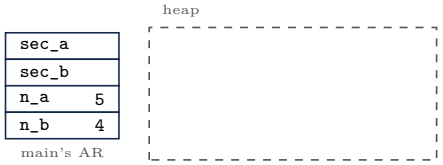
```
    student *sec_a, *sec_b;
    int n_a, n_b;
    sec_a = read_roster("roster.txt", &n_a);
    sec_b = read_roster("roster_b.txt", &n_b);
```

- Bytes asked for, `*n * sizeof(student)`: _____ for Section A, _____ for B
- `main`, after printing both:

  _____
- Section A now prints

  _____
- `ftranspose_heap.c`, the transpose, the same way: `m * n` ints once `m` and `n` are read; element (r, c) is `in_matrix[r * n + c]`

Draw `main`'s AR and the heap:



Losing a Block: Memory Leak

```
ptr = (int *)malloc(sizeof(int));
*ptr = 5;
ptr_2 = (int *)malloc(sizeof(int));
*ptr_2 = 6;
ptr = ptr_2;
printf("*ptr = %d, *ptr_2 = %d\n", *ptr, *ptr_2);
free(ptr_2);
```

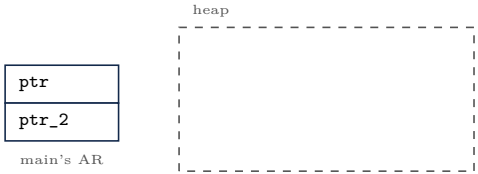
Output: `*ptr = 6, *ptr_2 = 6`

`leaks -atExit - ./leak` reports _____ leak

Add `free(ptr); free(ptr_2);` at the end instead:

_____

Draw the heap after `ptr = ptr_2`:



A **memory leak**: a block that is still allocated but that no pointer in the program reaches. It stays taken until the program ends.

The fix: _____ before `ptr = ptr_2;`

One free per malloc, through any pointer that still holds the address.

After free: the Dangling Pointer

```
ptr = (int *)malloc(sizeof(int));
if (ptr == NULL)
    return 1;
*ptr = 10;
free(ptr);
/* ptr dangles: the block is not ours */
printf("%d\n", *ptr);  /* undefined */
ptr = NULL;            /* the habit */
free(ptr);             /* does nothing */
printf("%d\n", *ptr);  /* seg fault */
```

One run, on a Mac:

```
_______
_____________________
```

The second `printf` is a **segmentation fault**: the operating system stops the program at that line.

- After `free(ptr)`, `ptr` still holds the address: a **dangling pointer**. The block is the allocator's again, and may already be in use.
- Reading or writing through it is _____
- `free(NULL)` does nothing, so a pointer set to `NULL` after `free` is safe to free again
- The opener's `sec_a` was dangling too: the AR that held its block had been popped

Set a pointer to NULL after free. A NULL dereference stops the program where the bug is.

calloc: a Zeroed Block

```
a = (int *)malloc(N * sizeof(int));
if (a == NULL)
    return 1;
for (i = 0; i < N; i++)
    a[i] = 7;
free(a);   /* the block goes back */

a = (int *)malloc(N * sizeof(int));
b = (int *)calloc(N, sizeof(int));
```

Then it prints the first five ints of each. On this Mac:

```
malloc: _____
calloc: _____
```

`N` is 50000. The 200,000-byte block came back as `free` left it. Small freed blocks on this Mac come back zeroed; the C standard promises nothing for `malloc`.

`void *calloc(size_t n_items, size_t item_size)`

- a block of _____ bytes, every byte set to zero
- otherwise as `malloc`: `NULL` on failure, one `free` for the whole block
- `char *ptr2 = (char *)calloc(100, sizeof(char));` gives _____ bytes of `'\0'`: already _____

Use calloc when the code reads from the block before it has written every byte.

realloc: Resize, and Maybe Move

```
ptr2 = (char *)calloc(100, sizeof(char));
if (ptr2 == NULL)
    return 1;
strncpy(ptr2, "Example using calloc", 100);
printf("%s, at %p\n", ptr2, (void *)ptr2);

ptr3 = (char *)realloc(ptr2, 200 * sizeof(char));
if (ptr3 == NULL) {
    free(ptr2);
    return 1;
}
printf("%s, at %p\n", ptr3, (void *)ptr3);
free(ptr3);
```

One run on a Mac; the addresses change from run to run:

```
Example using calloc, at 0x1033659c0
Example using calloc, at _________________
```

`void *realloc(void *ptr, size_t size)`

- resizes the block at `ptr` to `size` bytes and returns its address
- room behind the block: it grows in place, same address
- no room: a new block of _____ bytes; the old _____ bytes are copied over; the old block is _____
- `ptr2` afterwards: _____
- `free(ptr3)` releases _____ bytes. `free(ptr2)` as well? ____
- `ptr == NULL`: same as _____. `size == 0`: implementation specific

Keep the old pointer until realloc has succeeded: on failure the old block is still yours.

In-Class Problem: One Record, and Freeing Them All

`typedef struct studentStruct { char *NAME; int UIN; float GPA; } student;`

`sizeof(student)` = _____

```
void init_record(student *s) {
    /* 100-char block; copy "To be set" in */




    /* UIN -1, GPA 0.0 */



}
```

```
void free_all(student *records, int n) {
    int i;
    /* every NAME block first */




    /* then the records */


}
```

`init_record` asks for _____ bytes. `init_record(&records[i])` passes _____, so the function fills record `i` in place.

In-Class Problem: 200 Records, Then 400

```
int main() {
    student *records, *grown;
    int i;

    /* 1. a block of 200 records; NULL check */




    /* 2. initialize all 200 */
```

```
    /* 3. grow to 400: realloc, NULL check, */
    /*    records = grown, init 200..399    */




    /* 4. free everything */


    return 0;
}
```

After step 2, blocks on the heap: _____ bytes asked for: _____ After step 4: _____

The Heap After Step 3

Draw `records` and everything it reaches:



- Blocks on the heap: _____; bytes asked for: _____
- Calls of `free` in step 4: _____
- `records.c` prints, on this Mac:

```
records[199]: To be set -1 0.0
the block moved
records[399]: To be set -1 0.0
```

Whether the block moves depends on the allocator; the 200 records in it are copied either way.

`leaks -atExit - ./records`:

```
_____
```

Summary & What's Next

Today

- **The problem**: a local array dies with its AR; the second call of `read_roster` built its AR over the first roster
- **The heap**: memory the program asks for by size while it runs and keeps until it frees it; it sits between the global data and the run-time stack, and the two grow toward each other
- **malloc and free**: `(student *)malloc(n * sizeof(student))`, `NULL` on failure, `free` returns the whole block; memory leak, double free, dangling pointer, segmentation fault
- **calloc** zeroes the block; **realloc** resizes it, copies the contents to a new block when it must move, and frees the old one
- **200 records**: one block of structs plus one block per NAME, grown to 400 with `realloc`, freed NAMEs first; `leaks` reports 0

Next lecture: linked lists.

Code from today: `roster_stack.c`, `roster_heap.c` (skeleton `roster_heap_skel.c`; both read `roster.txt` and `roster_b.txt`), `malloc_int.c`, `leak.c`, `double_free.c`, `after_free.c`, `calloc_zero.c`, `realloc_move.c`, `records.c` (skeleton `records_skel.c`), `ftranspose_heap.c` (reads `in_matrix.txt`).